Predicting Post-Graduation Earnings with Machine Learning

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The Challenge

- U.S. student debt > $1.7 trillion
- Students choose colleges with limited ROI data

Core Question

→ Does paying more for college lead to higher earnings?

Our Goal

- Use machine learning to predict median earnings (10 years after enrollment)
- Identify which college factors actually matter

Approach and Models

How We Built It

- **Dataset**: 950 U.S. colleges (College Scorecard 2023)
- **Features**: tuition, SAT scores, debt, repayment rates, demographics

Models

- Linear Regression (baseline)
- Feedforward Neural Network (FFN)

Pipeline

User Data → Model → Salary Prediction → AI Career Advice

Results and Impact

Key Findings

- FFN performed best:
  R^2: 0.626 vs 0.587
- **Top Predictor:** SAT Math Score

  → Stronger than tuition cost
- **Main Insight:** Academic selectivity > price

Impact

- Helps students make smarter college decisions
- Built interactive tool with real-time predictions + AI guidance